Paleomagnetism of the Spanish River Alkalic Rock–Carbonatite Complex, Twps. 107 and 108, Ontario[1]

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The ijolite series and carbonatite from the Spanish River complex, Townships 107 and 108, Ontario, intruded 1790 ± 100 m.y. ago, are normally magnetized, yielding a pole position at 37° N, 96° W (dp = 24, dm = 25). Spatially associated dikes and fenite are reversely magnetized yielding a pole at 10° S, 136° W (dp = 11, dm = 21). The latter magnetization may be related to an earlier pole unmodified by the intrusion or to a minor thermal episode subsequent to the main intrusive event.

Les series d'ijolites et la carbonatite du complexe de la rivière Espagnole-cantons 107 et 108 Ontario—intrudés il y a 1790 ± 100 m.a. sont normalement magnétisés et donnent la position de pôle suivante: 37° N 96° W (dp = 24, dm = 25). Des dykes et des fénites associés spatialement ont un magnétisme inversé et donnent la position de pôle suivante: 10° S, 136° W (dp = 11, dm = 21). Cette dernière magnétisation peut être reliée à un pôle antérieur non modifié par l'intrusion ou à un épisode thermal mineur subséquent par rapport à l'événement intrusif majeur. [Traduit par le journal]

Introduction

The carbonatite–alkalic rock complex near the 'elbow' in the Spanish River, Townships 107 and 108, Ontario, is a very poorly exposed elliptical intrusion into Archean granitic rocks. Trenching and drilling by Union Carbide Exploration Co. (Fig. 1) have provided fresh material for magnetic and petrological studies. Much of the trenched material has weathered rapidly so that some digging was necessary to obtain fresh drill core for oriented samples. Sampling difficulties caused by weathering resulted in a small number of samples and large uncertainties in magnetization directions.

Each group of trenches or outcrop was treated as a separate site, but within most sites, wide variations of rock type have yielded different magnetic characteristics and these have been treated as sub-sites (Fig. 1, Table 1). Drill cores were oriented in the field by suncompass using standard paleomagnetic techniques. Angles should be accurate to within two degrees.

Geology and Age

The Spanish River complex is composed of rocks of the ijolite suite and carbonatite intruded into Archean granite and gneissic rocks. Minor syenite, nepheline syenite, and biotitic (in part vermiculite) rocks occur within the complex but their relationship to the ijolite suite and carbonatite is not clear because of the poor exposure. There is no petrographic evidence for metamorphism of the rock types and no apparent rotation of the intrusion since its emplacement. The complicated intermingling of rock types typical of carbonatite complexes is evident in a 1736 ft (529.5 m) hole collared at Site C and drilled into a magnetic anomaly by Union Carbide Exploration Co. (Watkinson and Grimes 1974). The drill hole intersects about 1066 ft (325 m) of fenitized country rock intruded by about twenty narrow lamprophyric dikes in the first 500 ft (152.5 m) and many more carbonatite dikes in the range 500–1066 ft (152.5–325 m). Within the contact of the main intrusion with the fenite, 450 ft (137 m) of approximately equal amounts of carbonatite and carbonate-bearing, pyroxene-rich members of the ijolite suite and biotite-rich rocks occur. Carbonatite predominates to the end of the hole.

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TABLE 1. Sub-site rock types and direction statistics after cleaning in a peak alternating field of 200 oe

Rock type	Sub-site	N	D	I	θ_{63}
Alkalic intrusion and associated lamprophyre					
Lamprophyre	C2	2	006	+79	22
Leucocratic feldspathic ijolite	D1	4	254	+58	17
Biotite-calcite ijolite	D2	2	193	+46	4
Fine-grained ijolite	D3	2	316	+78	24
Fine-grained biotite ijolite	D4	3	080	+78	18
Olivine-phlogopite-magnetite-carbonatite	E1	3	071	+69	19
Biotite-calcite ijolite	E2	3	280	+75	32
Coarse-grained cancrinite ijolite	F3	2	168	+57	32
Fenitized granite and dikes					
Calcite-bearing, altered diabase dike	A1	3	055	−61	30
Calcite-bearing, altered felsite dike	A2	3	026	−08	9
Fenite	B	4	067	−24	7
Fenite	C1	4	074	−23	45
Random sites ($\theta_{63} < 45°$)					
Biotite-calcite-ijolite	D5	3	021	+16	49
Olivine-phlogopite-magnetite-carbonatite	E3	2	154	+70	—
Coarse-grained ijolite	E4	2	069	+17	—
Coarse-grained cancrinite ijolite	F2	2	292	+30	—
Carbonatite dike	G1	3	201	+30	—
Fenite	G1	3	063	−10	55
Fenite	G3	3	112	−15	—

N is the number of separately oriented cores.
D is the declination east of true north.
I is the inclination, positive downwards.
θ_{63} is the circular standard deviation.

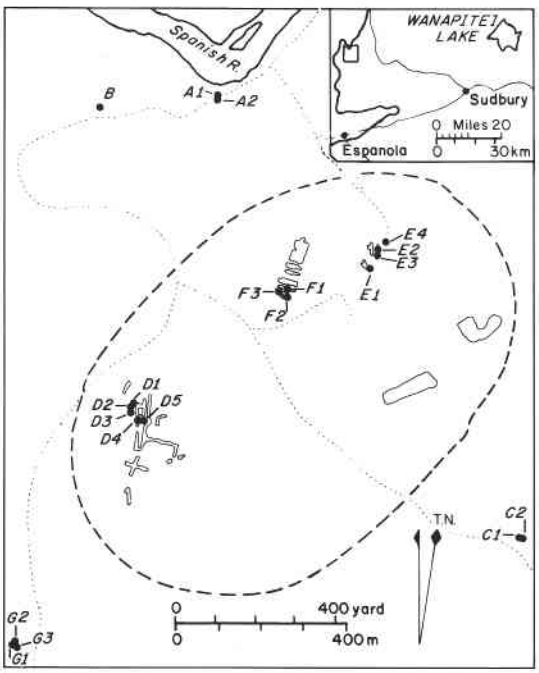


FIG. 1. Map showing the palaeomagnetic sampling sites and trenches in the Spanish River complex.

A potassium–argon age on biotite from a trench was reported to be 1560 m.y. by Gittins *et al.* (1967). Subsequent radiometric dating of fresh micas separated from drill core has revealed a wide range of K–Ar biotite ages. Data from drill core of the main intrusion give a minimum age near the contact of 1400 m.y. and a progressive increase to about 1800 m.y. well within the intrusion (Bell *et al.* 1974). This extremely large range prompted them to do a whole rock Rb–Sr isochron study of drill core; this gave a preliminary value of 1790 ± 100 m.y. (using $\lambda = 1.47 \times 10^{-11}$ y).

Paleomagnetic Results

In conjunction with the sampling for petrographic and age work, oriented core samples for paleomagnetism were taken from rocks of the ijolite series from trenches within the intrusion and from fenite, lamprophyre, and dikes nearby.

The scattered NRM directions, both from the alkalic intrusion and the surrounding rocks separate into two distinct groups after magnetic

cleaning in 200 oe (Fig. 2). Two factors probably dominate in causing dispersion of the NRM directions. Firstly, some of these alkalic rocks seem to be characterized by large grains of magnetite that are quite unstable magnetically. Sub-sites such as E3 (Fig. 3) appear to acquire large, low coercivity components such that the original direction is completely masked. Secondly, a tendency towards streaking between the two mean directions obtained from NRM caused by specimens containing a component from both the earlier and later events, appears to have been eliminated after 200 oe cleaning (Fig. 2). It may be presumed that these two directions represent two different ages, and the later event may be partly imprinted on rocks that were formed (and magnetized) earlier. Rocks that contain magnetic components from both ages will, of course, yield a direction between those of the two separate directions, being the vector sum of the two components

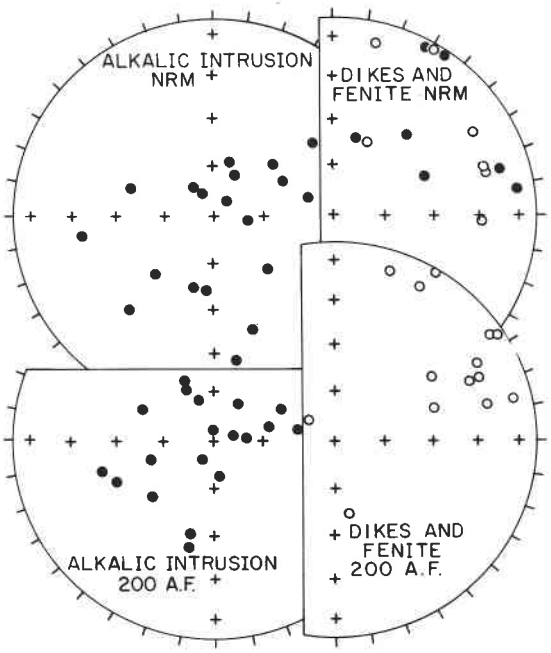


FIG. 2. Equal area stereograms showing directions from sites in which $\theta_{63} < 45°$. North seeking directions are plotted as open circles on the upper hemisphere, and solid circles on the lower hemisphere. The upper stereograms show sample directions from the intrusion, and the surrounding fenite and dikes respectively; the lower ones show the directions from the same samples after magnetic cleaning in a peak field of 200 oe.

present. Evidence both from intrusive relationships, and from the positions of the two poles relative to the North American polar wandering curve, indicates strongly but not indisputably that the direction from the alkalic rocks is the older of the two.

During alternating field (AF) demagnetization specimens from some sites (*e.g.*, E4 and F2, Fig. 3) acquired components in higher cleaning fields (*i.e.* 400 oe) comparable in magnitude to those of the NRM but unrelated in direction to them. The reason for this is not clear but it appears to be an effect of AF cleaning of low coercivity rocks, and may be related to the rotational remanent magnetization (Wilson and Lomax 1972). Thus all specimens were cleaned in an alternating field of peak value 200 oe, a field judged high enough to eliminate low-coercivity components and low enough to avoid the high field 'rotational effects'.

Table 1 shows the very wide range of within sub-site dispersion. Because of the low sampling density, caused by difficulties presented by the weathering of the rock, and the high dispersion at many sites, the usual criterion of Watson (1956) was not used for eliminating random sites. Instead sub-sites were considered to be random, and therefore excluded from the mean calculations, if the circular standard deviation (θ_{63} of Irving 1964 and Table 1) was greater than 45.

Table 2 shows the mean directions both of the alkalic complex, and of the dikes and fenite, calculated both using sub-sites and samples from nonrandom sub-sites as units. Although errors are quite large the two groups are significantly different. The directions indicate that the alkalic complex is normally magnetized, whereas the dikes and fenite are reversely magnetized; clearly there has been a polar shift between the two events.

Conclusions

The Spanish River alkalic rocks became normally magnetized in a high latitude field at the time of intrusion, and although subsequent partial remagnetization has altered the direction in many specimens, the approximate direction of the field at the time of cooling can be obtained from magnetically cleaned samples. These give a pole position in the center of

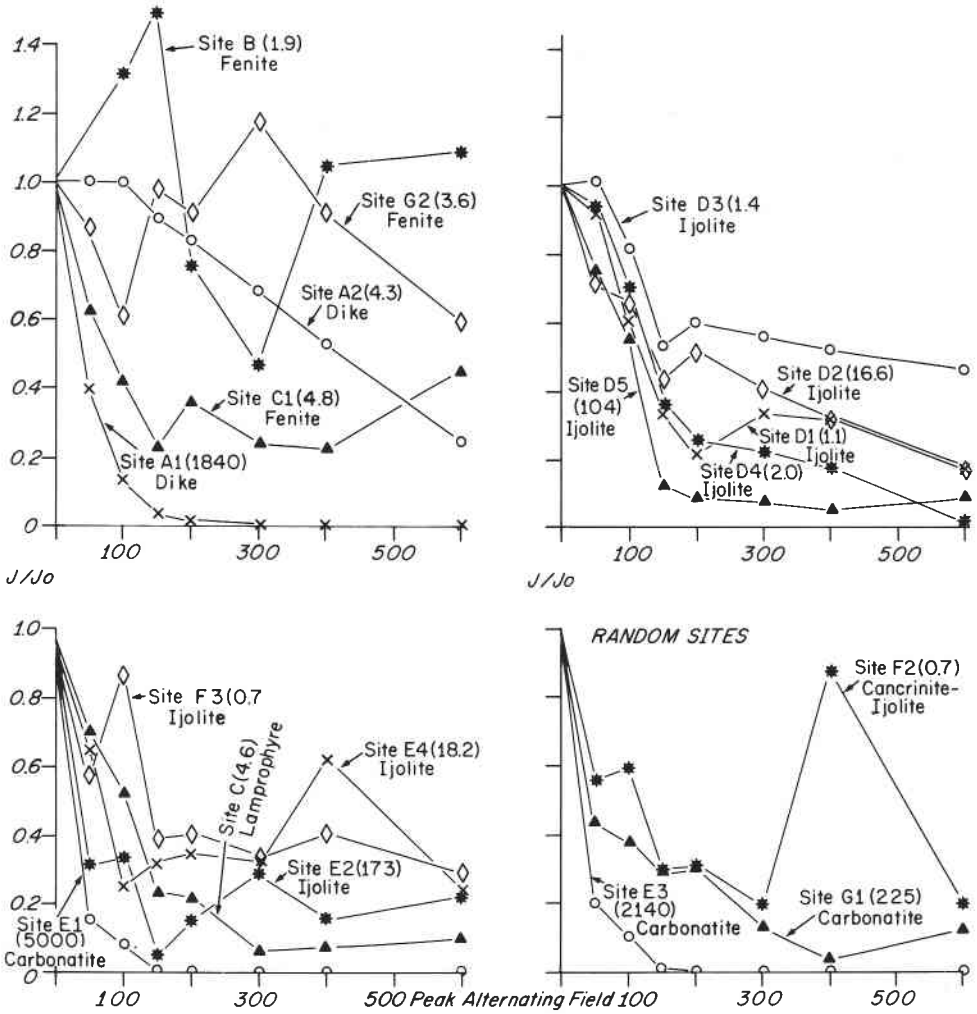


FIG. 3. Normalized alternating field demagnetization curves of a specimen from each subsite. The figures in brackets give the initial intensity of the specimen in e.m.u. $\times 10^{-6}$.

TABLE 2. Group statistics from samples magnetically cleaned in 200 oe

	N	D	I	R	k	α	Pole lat.	Pole long.	dp	dm
Alkalic complex										
Sub-sites	8	205	+83	7.30	10	18	34N	89W	35	36
Samples	21	234	+83	18.30	7	13	37N	96W	24	25
Fenitized granite and dikes										
Sub-sites	4	055	−30	3.60	8	36	10S	135W	22	40
Samples	14	056	−29	11.59	5	19	10S	136W	11	21

The symbols have the same significance as in Table 1.
dp is the semi-axis of the oval of confidence along the great circle passing through the formation, at a probability P = 0.95, and dm is the semi-axis perpendicular to dp.
k is Fisher's (1953) best estimate of precision.
α is the half-angle of the cone of confidence at a probability P = 0.95.

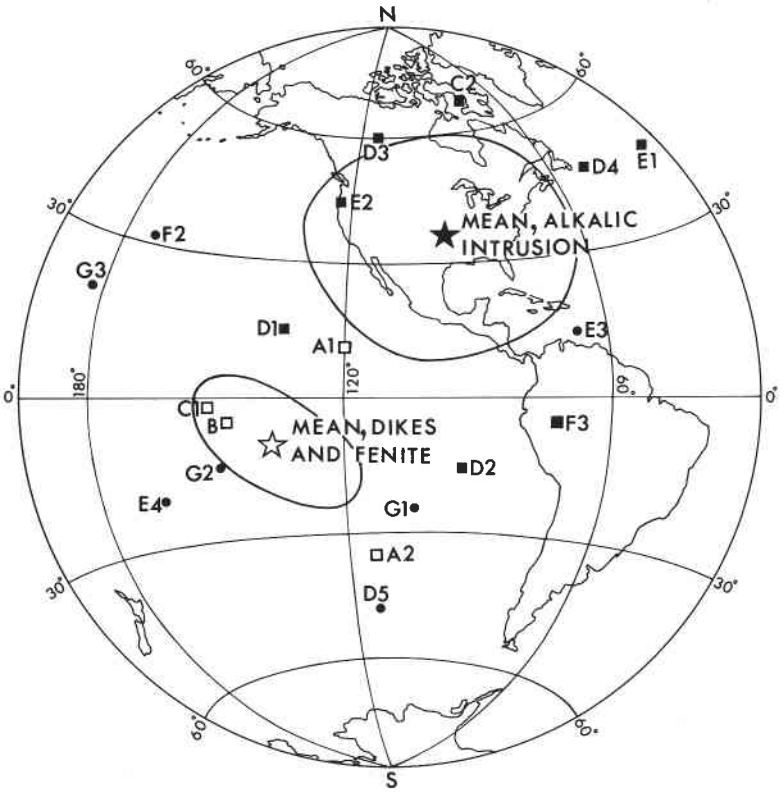


FIG. 4. An equal area projection of the western hemisphere of the globe (160E–20W) showing the pole positions of the sub-sites. Those from the main intrusion are shown as solid squares, and those from dikes and fenite by open squares. Those for which $\theta_{63} < 45°$ after cleaning are shown as solid circles. The means are shown respectively as a solid and open star, and the ovals of confidence, using samples as units and a probability P = 0.95, are given.

North America (Fig. 4). The main intrusive rocks presumably acquired this magnetization 1790 ± 100 m.y. ago.

The Spanish River pole (37° N, 96° W) is close to that for the Otish Gabbro (35° N, 107° W), (Fahrig and Chown 1973), which is of similar age or somewhat older and from the eastern part of the Canadian Shield. On the other hand, poles from the Nonacho Sediments, (13° N, 086° W), Sparrow Dikes, (12° N, 069° W) (McGlynn *et al.* 1974), the Et-Then Group, (01° S, 048° W) (Irving *et al.* 1972), the Dubawnt Group (07° N, 083° W) (Park *et al.* 1973), and the Martin Formation (09° S, 072° W) (Evans and Bingham 1973) from the western part of the Canadian Shield that probably span the time of formation of the Spanish River alkalic intrusion, all lie well to the southeast. This divergence of eastern and western shield poles of similar age suggests the possibility that relative movement between two blocks may have taken place after the rocks in them became magnetized. Uncertainties in the data make this no more than a speculation, but it points to the need for data from restricted Precambrian areas, and to the danger of drawing polar wandering curves through poles from Precambrian rocks that are widely separated in space, even though today they form one continental mass.

The dikes and fenite sampled near the intrusion were reversely magnetized in a low latitude field at some other time. They yield a pole position in the south-central Pacific Ocean. The position of this latter pole is consistent with those from Canadian Shield rocks em-

placed 1375–1475 m.y. ago (Robertson 1973), an age similar to the youngest K–Ar biotite age for rocks of the complex. However, there are no data on the 'age' of the dikes or of the fenite; the pole for the rocks outside the intrusion could conceivably be a much earlier pole that was not modified during the intrusion of the main complex.

Acknowledgments

We wish to thank Dr. Jaroslav Hak for his help in sampling during the height of the blackfly season, and Ted Irving and Keith Bell for critically reading the manuscript. Part of this study was financed by NRC grant A7874 to Watkinson.

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